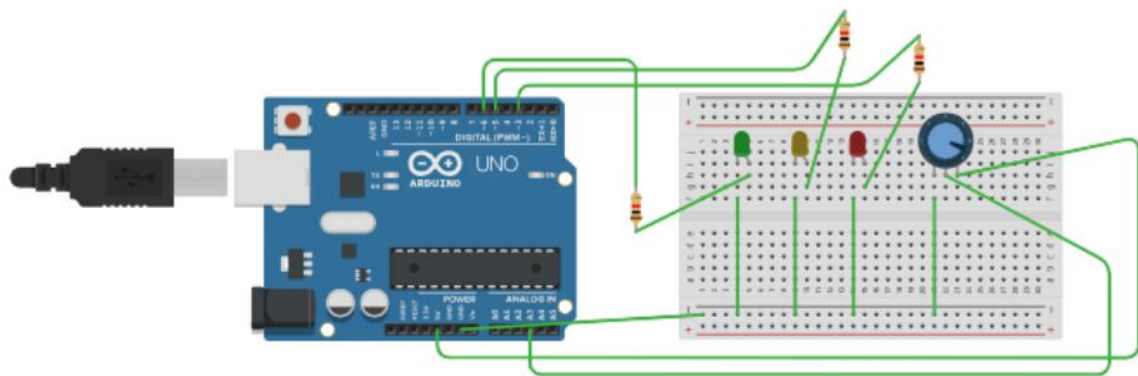


Robocon- Tinkercad Task-3.1

Circuit diagram and code explanation

Task- connect the potentiometer and three LEDs to the Arduino and write a program that reads the value of the potentiometer and controls the LEDs according to the given conditions

Circuit:



Explanation:

Components used— Arduino UNO, breadboard, potentiometer, 3 LEDs, 3 resistors, jumper wires

Code logic—

- We take the input from potentiometer and store it in a variable(input) using `analogRead()` function

- In order to give the output from a digital pin we convert the input value into valid output values using map function
- Depending on the potentiometer value we use if else condition to execute different conditions
 - If potentiometer value is in range of 0-340 green LED glows rest are off
 - For range of 340-680 yellow LED is on rest are off
 - For the range of 681-1023 red LED glows rest are off
- We print the potentiometer value and brightness value using Serial.print for all cases